ME 32301 Fall 2025

Basic Course Information

Course Title

ME 32301 — Mechanics of Materials Laboratory

Course Instructors

Prof. Tina Han
Email: han547@purdue.edu
Office Hours: by request

Prof. Zachary Harbin
Email: harbinz@purdue.edu
Office Hours: by request

Lecture Sessions

Section	Time	Location
25	Thursday 16:30-17:20	BHEE 129
26	Friday 9:30-10:20	BHEE 129

Lab Sessions

Location: ME 1003

Time	Monday	Tuesday	Wednesday	Thursday	Friday
07:30-09:20				16	
09:30-11:20	2	7	12	17	22
11:30-13:20	3	8	13	18	23
13:30-15:20	4	9	14	19	24
15:30-17:20	5	10	15	20	

Course TAs

Jackson Cruise	jcruise@purdue.edu	Tejas Kulkarni	kulkar72@purdue.edu
Johnson Olatunde	jolatund@purdue.edu	Uday Rade	urade@purdue.edu
Shayak Chatterjee	chatte26@purdue.edu	Zackary Van Zante	zvanzant@purdue.edu
Panagiotis Papageorgiou	ppapageo@purdue.edu	Yeongun Ki	yki@purdue.edu
Catarina Jancso Chapelain	cjancsoc@purdue.edu	Joel Francis	franc142@purdue.edu
Jose Miguel Junco Fresan	jjuncofr@purdue.edu	Charles Dicke	cdicke@purdue.edu

TA Office hours

The TA office hour schedule below pertains only weeks with Virtual Lab, with office hours held in ME 2138. During the weeks with In-Person Lab, TAs will be in ME 1003, with the lab session schedule listed previously. *Students are welcome to stop by the lab and ask questions as long as it does not interfere with the lab instruction. Students can go to any of the course TAs for questions, not limited to the TA of any specific lab section.*

Monday	11:30-17:20
Tuesday	9:30-17:20
Wednesday	9:30-17:20
Thursday	9:30-15:20
Friday	9:30-17:20

Course Description and Objectives

Learning Objectives: The ME32301 Lab course is structured to allow students to apply theoretical knowledge in practical settings and develop proficiency in both virtual and in-person lab environments. The objectives are categorized under virtual and in-person labs with a focus on computer modeling, simulation, hands-on skills, error analysis and technical objectives matching the theoretical course ME 323.

- **Virtual Labs:** In the virtual labs, students will develop proficiency in computer simulations and modeling. They will understand the fundamental principles behind the simulations and models, and will be able to analyze and interpret simulation data to make meaningful conclusions.
- **In-Person Labs:** In the in-person labs, students will gain hands-on experience with equipment for characterizing and processing solid materials. They will understand the theoretical principles in a real-world context and will develop skills in troubleshooting and problem-solving in a live lab setting.
- **Error Analysis:** With regards to error analysis, students will develop skills in conducting error analysis through Microsoft Excel and MathWorks MATLAB. They will understand and apply linear and nonlinear parameter estimation, and will apply bootstrapping to determine confidence intervals in MATLAB for linear and nonlinear problems.

Technical Objectives: By participating in this course, students will solidify their understanding of core mechanics of materials principles via direct, practical application. They will enhance their problem-solving abilities, honing skills that can be broadly applied to numerous mechanical engineering scenarios. Additionally, students will cultivate their proficiency in utilizing modern engineering software and equipment to tackle complex issues, thereby fostering a more practical and intuitive approach to engineering design and material selection.

Course Textbook

Required: *ME32301 Lab Manual* to be purchased at University Bookstore.
Optional: *ME323 Lecturebook* to be purchased at University Bookstore.

Course Management

- **Brightspace** will be used to host all the course material including handouts, homework questions and solutions, videos and etc.
- **iClicker** is used for in-lecture quizzes.
- **Gradescope** is used for all memo submission and other asynchronized quizzes.
- **CATME** is used for peer evaluation.

Copyright Materials

Please note that the ME 32301 Lab Manual, ME 323 Lecturebook, quizzes, etc. are copyrighted materials and should not be sold, bartered to others, or posted on-line without the expressed written consent of the authors. Similarly, notes taken in class are considered to be "derivative works" of the instructor's presentations and materials and likewise should not be sold or bartered or posted on the internet without consent. Students are permitted to use their notes, assignments, quizzes etc. for individual and/or group study or other non-commercial purposes reasonably rising up from enrollment in the course or the University generally.

Attendance Policy

Lecture/lab attendance is required, and lateness will be noted and accounted for in the grading. Students would not be allowed to join the in-person lab (ME 1003) after 10 minutes of the lab starting. An unexcused absence from an in-person lab session will result in the student being removed from their team and not being allowed to submit the corresponding lab memo. An unexcused absence from a lecture will result in the student missing the corresponding pre-lab quiz.
In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted on the course Brightspace page or communicated through Purdue email. You are expected to check and reply to your @purdue.edu email on daily basis.
Only the instructor can excuse a student from a course requirement or responsibility. When conflicts can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor of the situation as far in advance as possible so that alternate arrangements may be made. For unanticipated or emergency conflict, when advance notification to an instructor is not possible, the student should contact

the instructor as soon as possible by email. For cases that fall under excused absence regulations, you or your representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or emergent medical care. The processes are detailed, so plan ahead.

Course Grading

Your course grade be will based on a straight grading scale: 97-100% A+; 93-97% A; 90-93% A-; 87-90% B+; 83-87% B; 80-83% B-; 77-80% C+; 73-77% C; 70-73% C-; 67-70% D+; 63-67% D; 60-63% D-; <60% F.

- **Lab Memos [88%]**: There will be a total of 8 Lab Memos submitted and graded, with the lowest 1 being dropped. Each lab memo will be scaled based on the student's CATME grade for that lab. All lab memos are due at 11:59 PM on Fridays as shown in the schedule. A late penalty of 30% will be applied if the submission is submitted within 48 hours of the original due date. No further extension will be granted unless agreed upon beforehand. Please review your submission after it has been uploaded onto Gradescope to ensure that all work has been properly submitted and pages are properly assigned (not assigning pages properly will result in penalties in the lab memo). *Regrade request will be considered only if submitted within a week of when you first received your graded work. Your score might increase, decrease, or remain unchanged as a result of the regrade request.*

- **Quizzes [10%]**: There will be a total of 12 quizzes throughout the semester, including 10 pre-lab quizzes (iClicker) during the lecture, 1 syllabus quiz (Gradescope) and 1 Safety quiz (Gradescope). No make-up quiz will be given unless it is due to a university excused absence and was scheduled before the corresponding pre-lab lecture(s). To account for unforeseeable issues, the 2 quizzes with the lowest scores will be dropped.

- **Professionalism [2%]**: This course is a professional setting and students are required to act accordingly. Each lateness during in-person lab sessions will be noted and if there are 2 or more lateness, 1% of the final grade will be deducted. During lectures and labs, students should be paying attention and refraining from using their electronics for non-class-related purposes. Additionally, when communicating via email, students should be clear and concise, adhering to proper etiquette, and only reach out after they have made an effort to find answers on their own. Furthermore, regrade requests must include well-reasoned justifications rather than a mere desire for more points; grade grubbing will not be tolerated. Rest assured, the final grade calculation will account for rounding, so any requests for grade bumps or other forms of unfair treatment will not be entertained.

- **CATME**: CATME surveys are due on the Sunday of corresponding lab memo. To encourage teamwork, each CATME grade will be reflected on the respective lab memo as a scaling factor. CATME grades and comments will be released after each lab memo

so teammates have chances to give out feedback. Any CATME score lower than a 3 (average) will result in a factor smaller than 1 multiplied to the respective lab memo, with the lowest being 1 in CATME will result in a 60% scaling factor. The scaling factor is on a linear scale. In the case of a teammate not making any significant contribution, the rest of the team can make the decision to exclude the teammate from the submission, and the excluded teammate will receive a 0 for the lab. In those specific cases, a scaling factor greater than 1 could potentially be applied on a case-by-case basis when individual contributions are clearly documented to account for the extra work due to the not-contributing teammate. Not filling out CATME survey will result in a 60% scaling of the respective lab memo grade for the individual.

- **Extra Credit**: There will be extra credit opportunities throughout the semester. These opportunities will be announced either in-class or on Brightspace.

Academic Integrity

The instructor is a strong believer that the mission of the University can only be fulfilled in an environment of high academic integrity. Students are encouraged to review the University's policies on academic integrity, which is located at this website.
In addition, the Purdue University Honor Pledge is located here. Suspected cases of academic integrity violations will be reported to the Dean of Students.

Never forget:
As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.

Colaborative Work: Due to the collaborative nature of this course, all team members share responsibility for ensuring the integrity of the team submissions. This is because each person in the group is expected to actively contribute to and review the final submission to prevent academic dishonesty. The failure to catch plagiarism reflects a lack of diligence, and as such, all team members may face consequences for the unethical behavior of any individual within the group.
Usage of AI: Students should ensure that assignments submitted are original and based on their understanding. While AI can assist in research or provide general guidance, it should not produce work on behalf of the student. If a student collaborates with AI tools, they must specify the nature and extent of this collaboration in their submission. This includes providing details of the prompts used to generate the AI responses. Misuse of AI tools in coursework, which includes but is not limited to producing unoriginal work, uncited use of AI-generated content, or unauthorized assistance on assessments, will be considered a breach of academic integrity. Consequences will follow the Purdue's policies on academic dishonesty, which may include grade penalties, course failure, or more severe disciplinary actions.

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies.

Accessibility

Purdue University is committed to making learning experiences accessible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: `drc@purdue.edu` or by phone: 765-494-1247.

Mental Health/Wellness Statement

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc. sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours. The CAPS website also offers resources specific to situations such as COVID-19.

Disclaimer

This document is subject to change during the semester when deemed necessary.

Lab Safety

Lab safety is of paramount importance. The safety issues may include, but are not limited to, flying debris, pinching points, electrical safety, proximity to fast-moving equipment, working in confined space, etc. In addition, all the lab equipment and instruments are very expensive to replace. Please refer to the TA or instructor for any safety concerns. All personnel working in the lab should understand and commit to the following safety and behavior guidelines.

Every time you enter the lab, you need to sign in on the Lab notebook before starting the lab.

By signing in on the Lab notebook, you understand and commit to the following safety and behavior guidelines:

1. **Safety First:** Always wear safety goggles when in the laboratory. Ensure that you are familiar with the location and operation of safety equipment, including fire extinguishers and emergency exits.

2. **No Food or Drink:** Consuming food or beverages is strictly prohibited in the laboratory to ensure personal safety and the protection of equipment.

3. **Follow Instructions:** Always follow the laboratory procedures and protocols as instructed. If unsure about any step or procedure, seek clarification from teaching assistants.

4. **Respect Equipment:** Handle all laboratory equipment with care. Report any malfunction or breakage immediately to the instructor.

5. **No Unauthorized Experiments:** Only conduct experiments that have been approved and are part of the curriculum. Unauthorized experiments can be hazardous and may lead to disciplinary action.

6. **Clean Up:** After completing your work, clean your workspace, return all equipment to its designated place, and dispose of waste materials as instructed.

7. **Behavior:** Always maintain a respectful and professional demeanor. Horseplay, pranks, or any behavior that can distract or endanger others is strictly prohibited.

8. **Teamwork:** Collaborate effectively with your peers. Respect each team member's input and work together to achieve the best results in your experiments and projects.

9. **Report Accidents:** Immediately report any accidents or injuries to the instructor or lab supervisor, no matter how minor they may seem.

Failure to comply can result in removal from the lab and further disciplinary action.

Course Schedule

	PreLab Lec (TR-F)	Lab (M-F)	Due	OH
Week 01 08/25-08/29	Course Info & Lab 1: Tens. Test to Failure	–	08/29 –	N
Week 02 09/01-09/05	(Labor Day 9/02) Lab 2: 3D Printing Dogbones	Lab 1: Virtual Lab Tens. Test to Failure	09/05 2 Quiz	Y
Week 03 09/08-09/12	Memo Writing & Overleaf Support	Lab 2: In-Person (@ME1191) 3D Printing Dogbones	09/12 –	N
Week 04 09/15-09/19	Lab 3: Axial Loading of Dogbones	Lab 2: In-Person (@ME1191) 3D Printing Dogbones	09/19 Lab 1	N
Week 05 09/22-09/26	Lab 4: Torsion of Circular Shafts	Lab 3: In-Person Axial Loading of Dogbones	09/26 –	N
Week 06 09/29-10/03	Lab 5: Strain Gauge Rosettes	Lab 4: In-Person Torsion of Circular Shafts	10/03 Lab 2&3	N
Week 07 10/06-10/10	– –	Lab 5: Virtual Strain Gauge Rosettes	10/10 Lab 4	Y
Week 08 10/13-10/17	(October Break) Lab 6: Three Point Bending	Lab 5: Virtual Strain Gauge Rosettes	10/17 –	Y
Week 09 10/20-10/24	Lab 7: Uni. Ten. of Finite Width Plate	Lab 6: In-Person Three Point Bending	10/24 Lab 5	N
Week 10 10/27-10/31	Lab 8: Helical Extension Springs	Lab 7: Virtual Uni. Ten. of Finite Width Plate	10/31 Lab 6	Y
Week 11 11/03-11/07	Lab 9: Design of 3D Printed Parts	Lab 8: In-Person Helical Extension Springs	11/07 Lab 7	N
Week 12 11/10-11/14	– –	Lab 9: Virtual Design of 3D Printed Parts	11/14 Lab 8	Y
Week 13 11/17-11/21	– –	– –	11/21 Lab 9	Y
Week 14 11/24-11/28	(Thanksgiving Break) –	– –	11/28 –	N
Week 15 12/01-12/05	– –	– –	12/05 –	N
Week 16 12/08-12/12	– –	– –	12/12 –	N

Lab Session Teaching Assistants

There is no distinction between TA1 and TA2. Both TAs will be in the lab for 2 hours during the in-person lab session.

Sec No.	Day	Time	TA1	TA2
02	Monday	09:30–11:20	Joel Francis	Zackary Van Zante
03	Monday	11:30–13:20	Shayak Chatterjee	Tejas Kulkarni
04	Monday	13:30–15:20	Johnson Olatunde	Jose Junco Fresan
05	Monday	15:30–17:20	Johnson Olatunde	Panagiotis Papageorgiou
07	Tuesday	09:30–11:20	Uday Rade	Zackary Van Zante
08	Tuesday	11:30–13:20	Uday Rade	Catarina Jancso Chapelain
09	Tuesday	13:30–15:20	Tejas Kulkarni	Jackson Cruise
10	Tuesday	15:30–17:20	Jackson Cruise	Charles Dicke
12	Wednesday	09:30–11:20	Yeongun Ki	Panagiotis Papageorgiou
13	Wednesday	11:30–13:20	Shayak Chatterjee	Tejas Kulkarni
14	Wednesday	13:30–15:20	Johnson Olatunde	Yeongun Ki
15	Wednesday	15:30–17:20	Jose Junco Fresan	Yeongun Ki
16	Thursday	07:30–09:20	Charles Dicke	Panagiotis Papageorgiou
17	Thursday	09:30–11:20	Zackary Van Zante	Panagiotis Papageorgiou
18	Thursday	11:30–13:20	Johnson Olatunde	Zackary Van Zante
19	Thursday	13:30–15:20	Uday Rade	Joel Francis
20	Thursday	15:30–17:20	Uday Rade	Jose Junco Fresan
22	Friday	09:30–11:20	Tejas Kulkarni	Jackson Cruise
23	Friday	11:30–13:20	Shayak Chatterjee	Catarina Jancso Chapelain
24	Friday	13:30–15:20	Jackson Cruise	Catarina Jancso Chapelain